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Course Name: CIS 4374

Instructor: Professor Dobretsberger

Date: 9/10/26

Homework 1

Task 1:

Research: What already exists? Add the existing software landscape to your document, and how your application will be different from what already exists

My response:

A parking app that already exists is the ParkMobile app. The ParkMobile app allows a driver to find and pay for parking on the app. While looking at the app, I noticed that a driver only has access to the location of where they want to park, but not the parking spot when they have arrived at their destination. I believe that my application will be better from the Parkmobile app because my application will allow drivers to reserve a particular parking spot for when the driver arrives.

Task 2:

Start vision and scope section: Start with how the this project was acquired introduction of your (fictional company, and rough overview of what this project is about

My response:

The Smart Parking Platform was acquired after identifying the needs of finding parking for the drivers. The key goal of this project is to satisfy the users that are in need of finding available

parking, reserve parking spots, and paying for the parking space. This project will also benefit the parking operators by helping them keep track of the amount of parking spaces being used.

Task 3:

Deliver a draft for the SRS, including 15 user stories (minimum). This should grow over the course of the semester to completion.

1.) User registration and authentication

Actor: Driver

Purpose: The driver needs to create an account before reserving a parking spot.

Steps:

- The driver would need to create an account to log into the app (email, password, and vehicle information).
- The system would now send a verification code.
- The driver then uses the code to complete registration and log in.

2.) Real-time parking space availability

Actor: Driver

Purpose: Showing the drivers the available and occupied parking spots

Steps:

- The driver would need to select which parking garage that they would want to go to.
- The app will then check the real-time information from the sensors to see which parking spaces are available or occupied.

- The app will show the driver the available spots that are green and the occupied spots that are red.

3.) Interactive map displaying available parking locations

Actor: Driver

Purpose: The drivers are provided with a layout of the parking garage

Steps:

- The driver looks in the app to find the parking garage that he/she wants.
- The driver selects which floor level they would like to view
- The app shows where to drive and which parking spaces are available.

4.) Parking reservation capability

Actor: Driver

Purpose: The drivers are reserving a specific parking spot before arrival

Steps:

- The driver selects an open spot on the floor map.
- The driver clicks on "Reserve Spot."
- The system sets a 25 minute time period so no other drivers can claim it.

5.) Digital payment processing

Actor: Driver

Purpose: Allowing the drivers to park for their parking ticket.

Steps:

- The driver opened the payment checkout section
- The driver then selects a payment method
- The system processes the transaction and sends them an email confirmation code.

6.) Reservation history and receipts

Actor: Driver

Purpose: The driver is able to view past parking history and view past receipts.

Steps:

- The driver clicks on the "Reservation History" tab to review their past transactions.
- The system shows the driver the list of past parking visits.
- The driver can view any past visits and download their receipts.

7.) Administrative dashboard for parking operators

Actor: Parking Operators

Purpose: Parking Operators are able to manage their parking garage.

Steps:

- The Parking is able to log into the system
- They can select their registered facility
- The system shows all the data for the parking spaces, prices, and other facility utilities.

8.) Occupancy reporting and analytics

Actor: Parking Operator

Purpose: The parking Operators

Steps:

- The Parking operator views the "Analytics" section on the dashboard.

- The system will now display a chart showing the daily activity.
- The Parking operator can choose to view information at any time period (day, week, month, and year).

9.) Notifications and alerts for users

Actor: Driver

Purpose: Alerting the drivers on when their parking time is close to expiring.

Steps:

- Once the driver has reserved a parking spot, the system then starts a 25-minute countdown timer for the driver to arrive.
- The system then notifies the driver 25-minutes before the time expires.
- The driver has the option to extend the parking reservation.

10.) Integration with external mapping/navigation services

Actor: Driver

Purpose: The app provides the driver direction.

Steps:

- On the app the driver clicks on "Navigate" to know where the drivers are heading to.
- The app is then provided with a GPS so that the driver can know where the garage entrances are.
- Once the driver enters the garage, the app will then explain the directions on where to go and find the reserved parking spot

11.) Dynamic pricing rules/ events

Actor: Parking operator

Purpose: The parking prices increase during busy events

Steps:

- The parking operators when the parking prices increase or decrease

- The system monitors how occupied the parking garage gets during busy hours (busy events).
- The app will show the drivers the updated prices.

12.) Garage layout of the parking garage

Actor: Parking Operator

Purpose: Creating a layout of the parking garage for the drivers to view.

Steps:

- The operator creates a layout of the parking garage.
- The parking operator creates the levels, lanes, and parking spaces.
- The parking operator then links an overhead sensor to track the parking spots.

13.) Parking expiration time & session extension

Actor: Driver

Purpose: When the parking expiration is about to expire, the driver extends the reservation time.

Steps:

- The system will notify the driver that their parking time is about to expire.
- The driver has the decision to choose if he/she wants to add more time to the reservation.
- The driver would then need to pay a small fee for extra time.

14.) Automated exit and digital receipts

Actor: Driver and System

Purpose: The system ends the parking session by giving the driver a receipt and opening the gate.

Steps:

- Once the driver leaves the parking spot the overhead sensor will turn red
- The system has updated and has kept the spot available.
- The system will give the driver a receipt and open the exit gate for them.

15.) Filtered reservation for electric vehicle

Actor: Driver that owns an electric car

Purpose: Has a filtered layout for where the charging stalls are and allows the drivers to reserve it.

Steps:

- The driver goes to the app and clicks on the "EV Charging" filter tab.
- The app will then show the drivers where the EV charging stalls are located in the parking garage.
- The driver reserved an EV spot.

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Homework 2

Component 1: Driver's authentication and security

- Account registration & identity verification for the user
 - Build account registration (includes: email, password, vehicle information).
 - Send and verify the user's verification code.
 - Secure the user's password and login session.

Component 2: Parking availability and reservation

- Garage mapping & sensors
 - Build a real-time overhead sensor (green means the parking spot is available, and red means the spot is taken).
 - Display parking availability on each parking layout.
 - Provided a filter for electric vehicle (EV) charging parking spots.
- Reservation & parking sessions
 - Allow drivers to reserve a parking spot for 25 minutes.
 - Provided registration countdown and expiration alerts to notify the driver.
 - Allow drivers to extend their reservation for an additional fee.

Component 3: Payment to access official parking & automated gate access

- Payment
 - The driver pays with a credit card or any other digital wallet.
 - Generate and email the user's receipts.
 - Users are able to view payment history and download their receipts.

- Automated gate access
 - Connect an overhead exit sensor to detect when a vehicle leaves.
 - The exit gate automatically opens once a driver is done with their session.

Component 4: digital reports/ analytics and dynamic prices

- Digital reports
 - Create charts for parking occupancy by day, week, month, and year.
 - Allow drivers to view transaction history and download receipts.
 - Allowing the operators to download the summary reports of the parking garage revenue.
- Analytics & dynamic prices
 - Creating a dashboard to monitor and view parking availability
 - The system will automatically change parking prices during busy events.

Task 2: Draft timeline (Including Gantt chart)

Timeline:

Component	Tasks	Start	End	Duration
1- Driver's authentication and security	Build account registration	9/16/2026	9/20/2026	5
1.1.1	Send and verify the user's verification code	9/21/2026	9/23/2026	3
1.1.2	Secure the user's password and login session	9/23/2026	9/27/2026	5
Parking availability & reservation	Build a real-time overhead sensor	9/28/2026	10/3/2026	6
1.2.1	Display parking availability on each parking layout	10/4/2026	10/9/2026	6
1.2.2	Provide a filter for electric vehicle (EV) charging parking spots	10/10/2026	10/14/2026	5
1.2.3	Create 25-minute reservation feature	10/10/2026	10/17/2026	8

1.2.4	Add countdown & expiration alerts	10/15/2026	10/22/2026	8
1.2.5	Add reservation extension fee	10/23/2026	10/26/2026	4
Payment & automated gate access	The driver pays with a credit card or any other digital wallet.	10/25/2026	11/4/2026	11
1.3.1	Generate and email the user's receipts.	10/25/2026	11/4/2026	11
1.3.2.	View payment history and download receipts	11/5/2026	11/8/2026	4
1.3.3	Connect overhead exit sensor	11/9/2026	11/12/2026	4
1.3.4	Automatically open exit gate	11/13/2026	11/17/2026	5
Digital reports, analytic, and pricing	Create parking occupancy charts	11/13/2026	11/20/2026	8
1.4.1	Add transaction history & receipt downloads	11/16/2026	11/21/2026	6
1.4.2	Create parking garage revenue reports	11/18/2026	11/23/2026	6
1.4.3	Create parking availability dashboard	11/21/2026	11/30/2026	10
1.4.4	Add automatic dynamic pricing	11/24/2026	12/4/2026	11

Gantt Chart:

Smart Parking Platform

